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AN ALLEGED CASE OF "INFLECTIONAL CONTAMINATION": ON THE *I*-STEM INFLECTION OF LATIN *CIVIS*

1. Introduction

1. 1.

The centrality of Lat. *cīuis* for Roman and Italic culture tends to obscure the fact that the word incorporates a number of linguistic problems. Some of these problems could be described as fundamental, insofar as they involve questions about the root of the word itself, from the point of view of Proto-Indo-European.

1. 1. 1.

A fundamental formal problem concerns the presence or absence of a laryngeal at the end of the root. Thus, a short-vowel zero-grade form like Sanskrit *śivá-* 'auspicious, dear, friendly' points to a root without final laryngeal; but on the other hand, as noted by Lubotsky (1988: 94, cf. *EWAia* II: 640, 655), a root-final laryngeal seems indicated by the intonation of the Latvian full-grade form *siẽva* 'wife' (which, apart from the intonation and gender, corresponds to Skt. *śéva-*, with the same meaning as *śivá-*). But as interesting and important as this problem may be from the point of view of Indo-European, it has little importance for Latin; or, to put the matter differently, Lat. *cīuis* itself has nothing to contribute to this problem. As it happens, the development of laryngeal sequences of this kind in Latin is entirely unclear, as shown by Schrijver (1991:282-5) and as discussed more recently by Nussbaum (1997: 200).[1]

1. 1. 2.

An even more basic uncertainty concerns the root itself: which of the Proto-Indo-European roots with the shape **ĥeu̯-* or *ĥeu̯(H)-* is involved, and how

[1] As pointed out to me by S. Neri, Latv. *siẽva* could have the same derivational source identified by Nussbaum (1986: 14) for a series of Balto-Slavic feminine forms with lengthened grade, evidently derived from parallel masculines by a vṛddhi process, as in **u̯ōrnā-* (Li. *várna* etc.) 'raven' beside masc. **u̯orno-* (Li. *vařnas* etc.), cf. **kōru̯ā* 'cow' (Li. *kárvė*, SCr. *krȍva*, etc.), no doubt derived in the same way from a masc. **koru̯o-* (cf. Lat. *ceruos* 'stag'). Thus a Baltic **kēiu̯ā* 'wife' need have nothing at all to do with a root-final laryngeal (as is also suggested by the probable etymology of *siẽva* and related forms, discussed below [3.2.1.1.]).

should the meaning of that root be understood? It will be more convenient to return to this problem later (**3.2.1.1.**), after considering the meaning of Lat. *cīuis* and its cognates (**3.2.1.**).

1. 1. 3.

Beyond the root itself, another fundamental problem – and the focus of the present study – concerns the morphological structure of Lat. *cīuis*, and specifically its inflection as an *i*-stem. In fact, the Latin *i*-stem inflection of this word appears to be totally anomalous, from a historical point of view. What is normally assumed is an underlying thematic **ĥéi̯u̯o-*: compare Skt. *śéva-* (**1.1.1.**), as well as the Germanic **heiwa-* 'household' that appears, for example, in the first member of the Gothic compound *heiwafrauja* 'οἰκοδεσπότης'; and this presumed thematic **ĥéi̯u̯o-* should have resulted in a Latin *o*-stem †*cīuos*, not an *i*-stem *cīuis*.[1]

1. 2.

In order to pursue this morphological problem in more depth, we should consider approaches based on both inflectional and derivational morphology.

1. 2. 1.

There is no obvious solution from the point of view of inflectional morphology, since no close ties connect *o*-stem and *i*-stem inflection in Latin historically (unlike, for example, the close historical ties between *i*-stems and consonant stems). Outside Latin, one can find limited connections between *o*-stems and *i*-stems, as in the Sabellian *o*-stem genitive singular desinence (O. *-eís* etc.), imported from the *i*-stems. But this is a *uox nihili*, of no relevance for Lat. *cīuis*: there is no reason to suppose that the morphological peculiarity in question has anything to do with non-Latin Italic, much less with the genitive singular.

1. 2. 2.

Nor, despite some superficially relevant possibilities, is there an obvious solution based on derivational morphology. To be sure, *o*-stem and *i*-stem formations sometimes alternate in Latin synchronically; but these alternations generally involve certain well-defined derivational relationships, none of which has so far been identified in the putative pair consisting of a pre-

[1] The inflectional class of Oscan *ceus* (nom. sg., *Tab. Bant.* [ST Lu 1] 19) cannot be determined, and the word should thus be left aside. It is quite likely, in fact, to have been borrowed from Lat. *cīuis* in the first place, see Untermann 2000:395 and more recently Rix 2004: 499.

Latin **keiu̯os* beside attested Lat. *cīuis*. For example, the *i*-stem compound type *inermis, imberbis, immānis*, etc., beside simplex thematic *arma, barba*, Old Latin *mānus*, etc., is of no relevance, since there is no evidence that Lat. *cīuis* is decompositional. Nor does Lat. *cīuis* belong to any other previously-identified class of Latin secondary *i*-stems, such as the adjectival type *leuis, grauis*, etc., which, according to the traditional assumption, somehow continues old *u*-stems.[1] Similarly, some root nouns and other consonant-stems develop secondary *i*-inflection in Latin (cf. **1.2.1.**); but this is also a blind alley, since there is no evidence (and virtually no possibility) that *cīuis* derives from any sort of consonant-stem. In still other cases, the nature of the alternation is simply obscure, and therefore of no value for the problem at hand. Thus, for Lat. *fortis* 'strong', an original *o*-stem is often assumed, on the basis of the glosses *horctum* and *forctum* in Paulus ex Festo (91.14L). But there is much about this material (as well as possible related material) that remains unclear.[2] Nor is it even clear that the problem of adjectival *i*-stems is to be considered on the same footing as purely substantival forms like *cīuis*: traditionally, they are kept apart,[3] though we will have occasion to revisit this conception later.

1. 3.

In the end, therefore, we face a conundrum: the *i*-stem inflection of Lat. *cīuis* is unexpected, and yet it apparently cannot be motivated by means of hitherto identified processes of either inflectional or derivational morphology. As often in such desperate circumstances, the only recourse seems to be an appeal to so-called "contamination". In other words, we may hope to find a semantically related *i*-stem that has exerted sufficient pressure to induce the presumed original *o*-stem to change its inflectional class. This, in fact, is precisely the solution that is found in all Latin handbooks and now accepted as doctrine: the analogical source is said to be Lat. *hostis* '(public) enemy', which is indeed well-attested in paired collocations with *cīuis*, as

[1] See e.g. Leumann 1977: 346, Sihler 1995: 277, Meiser 1998: 120, 142.

[2] Thus, the *Forctibus* correlated with *Sanatibus*, in a fragment from the Laws of the Twelve Tables (Festus 474.27L), serves to complicate the case: as noted in Ernout-Meillet, *Forctes* (and perhaps even *o*-stem *horcto-*) may actually refer to a tribal name, the relationship of which to the adjective *fortis* remains unclear. Etymologically, the *t*-formant of *fortis* has been thought (thus explicitly e.g. Katz 1998b: 216n95) to imply a basis in a zero-grade *tó*-participle **dʰr̥gʰ-tó-* (: Av. *dərəzra-* 'firm', Lith. *dir̃žnas* 'tough', etc.; IEW 254, Kümmel 2000: 236n330), which would at least support an original thematic formation; but again, the morphological relationship between this presumed participle (possibly reflected in the gloss *horctum*) and the much better attested adjective *fortis* remains unexplained. (We will, however, return briefly to *fortis* below, 4.3.3.) Once again (cf. p. 140, n. 1): the Oscan comparative adverb *fortis* (*Tab. Bant.* [ST Lu 2] 12) can tell us nothing about its inflectional class, and this form, too, may well be loaned from Latin in any case (see Untermann 2000: 304f.).

[3] See, for the moment, Sihler 1995: 352 and Rasmussen 1987[1999]: 294ff., addressing the type Lat. *similis*, OIr. *samail* (vs. Gk. ὁμαλός) along similar lines.

often noted (see e.g. Walde-Hofmann [s.v. *cīuis*] and Ernout-Meillet [s.v. *hostis*]). In addition to the standard etymological dictionaries of Latin, the same solution is presented by numerous works on Latin, Italic, and Indo-European historical grammar.[1]

Analysis by contamination is a legitimate tool in the arsenal of historical linguistic methodology, given a number of documentable cases involving contamination; one cannot, therefore, claim that all such explanations are to be rejected in principle. There is, nevertheless, considerable methodological interest in constraining such analyses as far as possible, given the open-ended nature of alleged contamination patterns and their potential for abuse in historical explanations. In the present case, I hope to show that if a contamination has indeed taken place, its nature was somewhat more complex than has been granted so far; and more importantly, I will suggest that a more attractive solution to this problem may well involve derivational morphology after all, given recent work on certain derivational patterns affecting Indo-European thematic formations.

2. Problems with the traditional account

2. 1.

An immediate difficulty with the traditional account of Lat. *cīuis* consists in the total absence of any evidence, even indirect, pointing to the survival of a thematic **keu̯o-* into Latin (or, for that matter, Italic). After all, a contamination between PIE **ḱéu̯o-* and **gʰosti-* (yielding an *i*-stem **ḱeu̯i-*) could have occurred at any point from Proto-Indo-European onwards; but according to the traditional solution, this did not take place until shortly before the Latin period. Yet there is no trace in Latin of the thematic base form that is required for this analysis.

2. 2.

Nevertheless, the traditional account of Lat. *cīuis* by means of contamination has a certain superficial attractiveness, especially given the commonplace assumption that antonymic pairs (a notion to which we will return) are often subject to developments involving contamination. For Latin, the clearest instance of this sort – and one that on the surface shows much the same morphological development as the one claimed to underlie Lat. *cīuis* – appears in the adjective *hilaris* 'cheerful', regularly a third-declension form in the Classical language. But as a loan from Greek ἱλαρός, the older form, predictably, is an *o*-stem *hilarus*, which is still the normal form for Plautus.

[1] E.g. Leumann 1977: 343, Untermann 2000: 395, Mallory and Adams 1997: 214, 622, *EWAia* II: 654, among many others.

In fact, as discussed by R. Gerschner (2002: 170), even though the third-declension form cannot be entirely eliminated from the Plautine manuscript tradition, the two instances where it may occur in the text of Plautus (Po. 1367, Mo. 318) may actually reflect modernizations that have nothing to do with genuine Plautine usage, which may have been restricted to the *o*-stem form. The Classical form *hilaris*, then, is generally ascribed to pressure from its antonym *trīstis* 'sad, stern', both forms occurring naturally (and frequently) in paired collocations. This may not, however, exhaust possible sources for a third-declension *hilaris*: given the evidence from Plautus, which shows that the form *hilaris* arose well after the period of vowel-weakening, such a form can also be the product, in part, of influence from the surface stem-form that appears in the corresponding abstract *hilaritūdō*, which is freely used by Plautus (Cist. 54, Mil. 677, Rud. 421); cf. the parallel pattern in pairs like *suāuitūdō* (Bacch. 18, Stich. 755) beside *suāuis* (frequent in Plt.), not to mention the fact that a †*hilarotūdō*, of course, is an impossible formation for such a date.

2. 3.

For Latin, this example of lexical contamination and a number of others have recently been discussed by Bârlea and Bârlea (2001: 672f.), in a study of Latin antonyms. But all such cases show features that differentiate them from the traditional solution envisaged for Latin *cīuis*.

2. 3. 1.

The most obvious difference is that all such pairs are actually attested within the historical period, as distinct from the unattested thematic **keu̯o-* that is presumed to underlie the *i*-stem *cīuis* (**2.1.**). In fact, Latin words that seem to have undergone a contamination are mainly attested in late Classical and even post-Classical times. Of course, the mere existence of such forms lends some theoretical support to the possibility of a prehistoric instance of lexical contamination; but this possibility is further weakened by other considerations, such as the following.

2. 3. 2.

Documented cases of lexical contamination in Latin only rarely involve a "pure" change of inflectional class (as with OLat. *hilarus* replaced by Class. Lat. *hilaris*, and the presumed background of *cīuis*). Much more often, there is no change in inflection, but rather some other formal adjustment, typically involving the root vocalism: thus, for example, a Vulgar Lat. **greuis* 'heavy' is needed to account for certain Romance developments, replacing Class. Lat. *grauis*, under influence from *leuis* 'light'. Alternatively, we find an

adjustment in a suffix formant, as in Ecclesiastical Lat. *aeternālis* for Class. Lat. *aeternus*, after the paired form *mortālis*. In some cases, inflection does change, but only in a limited way: thus St. Augustine's *frīgor* (gen. *frīgōris*) 'cold' replaces Class. Lat. *frīgus* (gen. *frīgoris*) under contamination with masculine abstracts like *calor/calōris* 'heat' – but here, the new form remains a third-declension *r*-stem (or *s*-stem, in historical terms).

2. 3. 3.

Turning to semantics, we can also observe that most cases of contamination of the type under discussion involve highly marked antonymic pairings of rather basic vocabulary, as in the forms just cited: 'cheerful'/'sad', 'heavy'/'light', 'cold'/'heat', and so on. This is quite different from the complex societal notions inherent in Lat. *cīuis* and *hostis*: however one wishes to define them, these terms, while certainly contrasting, clearly do not reflect a simple antonymic couple.

2. 3. 4.

In fact, there is a semantic complication with this pair that seems never to have been acknowledged. Within the recoverable history of Latin, the *i*-stem (or, in Proto-Indo-European terms, the *ti*-stem) that underlies *hostis* preserves a meaning that is quite different from 'enemy' or 'stranger', or anything similar one might wish to set up as a semantic antonym of *cīuis*. I refer to a meaning that incorporates the notion of 'exchange', as visible in Lat. *hostia* 'sacrificial victim' [Enn., Plt.+] and in the verb *hostīre* 'to compensate, pay back' [Enn., Plt.+] – forms which, in my view, continue the inherited meaning of the Proto-Indo-European root **g^{h}es-* or **g^{h}as-*, as seen most clearly in the Old Hittite postposition *kāššaš* 'in exchange for' (< *'substitution').[1] This means that the pre-Lat. **χosti-* that was allegedly in contact with a pre-Lat. **keiu̯o-* will have covered a rather broad semantic range – much broader than simply 'enemy' or 'stranger', and prominently including the semantic field of "exchange"/"reciprocity", with little direct connection to Lat. *cīuis*. This fact, it seems to me, further undermines the plausibility of a simple "antonymic contamination" involving pre-Lat. **keiu̯o-* and **χosti-*.

2. 4.

It is also possible to consider lexical contaminations outside Latin, as in the discussions of contamination to be found in such standard handbooks of

[1] Differently on *hostia*, *hostīre* etc.: Eichner 2002, separating these forms from *hostis* (**g^{h}ós-ti-*) and assigning them to the root **g^{h}es-* underlying the PIE word for 'hand'. (I hope to return to this material in more detail elsewhere.)

historical linguistics as those of Anttila (1989: 76, 95), Hock (1991: 197ff.), Campbell (1999: 104f.), or Sihler (2000: 83ff.), to name only a few recent treatments in English. It is a commonplace, in such discussions, to emphasize semantic relationships involving either similarity or antonymy among the forms claimed to show contamination developments.[1] Yet none of these works cites a case that is precisely like the alleged Latin contamination under discussion ("**keiu̯o-* X *hostis* → *cīuis*"), in which an isolated lexical item undergoing influence by contamination simply ends up in an unrelated inflectional class. Thus, it seems that what we may call "antonymic inflectional contamination" (of the type generally assumed for Lat. *cīuis*, though with necessary reservations, in this case, about the "antonymy" involved; **2.3.4.**) is exceedingly rare – not just in Latin, but also from a purely typological perspective.[2]

2. 5.

We can conclude this part of the discussion by saying that it is legitimate to entertain doubts about the traditional method of accounting for the *i*-stem inflection of *cīuis* as the result of a simplistic "inflectional contamination". If contamination was involved, then the forces at work may have been more complex, and additional factors may also need to be considered.

3. Toward a new solution

3. 1.

In order to approach a more satisfactory account of the *i*-stem inflection of Lat. *cīuis*, let us begin by recalling a fundamental insight into Indo-European derivational morphology, itself going back to Brugmann, but developed more recently by the late Jochem Schindler.

3. 1. 1.

In his classic paper on the Sanskrit *cvi*-formations (1980), Schindler elucidated a derivational process whereby thematic nominal formations alternate

[1] On antonymic contamination, see the classic study by Malkiel (1951), with rich documentation from Vulgar Latin and Romance.

[2] Michael Weiss points out to me a possible instance of this sort in Umbrian, where (abl. sg.) **peři**/*peri* 'foot' can be interpreted as showing (unhistorical) *u*-stem inflection, under the influence of (abl. sg.) **mani**/*mani* 'hand'. But other interpretations of the ending are possible (see Untermann 2000: 522), and in the absence of any other attested case forms, a complete transfer of the word to *u*-stem declension is not demonstrable in any case. Still, if this example is accepted, it would involve a marked semantic pairing ('hand'/'foot') similar to those discussed above (**2.3.3.**). More or less comparable cases of "morphological cross-pollination" adduced by Sihler (2000: 84f.) likewise involve individual case forms and not the restructuring of entire paradigms, e.g. Hom. dat. pl. *υἱάσι* 'sons' after *πατράσι* 'fathers', etc.

with forms containing the vowel *-i-*, in such a way that the *-i-* replaces the thematic vowel. One of the clearest instances of this pattern appears in derivational pairs consisting of a thematic adjective beside an *i*-stem abstract or substantivization, as in the following examples:

PIE *$h_2ak̂ro$- (Gk. ἄκρος 'topmost')	*$h_2o/ek̂ri$- (Lat. *ocris* 'peak', Gk. ἄκρις 'hilltop')
Ved. *jírá-* 'swift'	*jíri-* 'rapids'
OCS *zŭlŭ* 'bad' ('evil', adj.)	*zŭlĭ* 'badness' ('evil', abstract)

This pattern is well-attested throughout Indo-European, most often in isolated forms, rather than as part of productive synchronic systems. This therefore reflects an archaic derivational relationship between *o*-stems and *i*-stems that is reconstructable for Proto-Indo-European itself.[1]

3. 1. 2.

Several such pairs have also been identified within Latin (see already Schindler 1980: 390): note the *o*-stem adjective *rauus* 'hoarse' [P.F. 355.3L] beside the *i*-stem abstract *rauis* 'hoarseness' [Plt.+]; similarly *gibber* (*-a, -um*) 'hunchbacked' beside (concretized abstract) *gibber* (gen. *gibberis*) 'hump (on the back)', and OLat. *tor(r)um* 'torridum, aridum' [Fest. 484.4L, P.F. 485.1L] beside (concretized abstract) *torris* 'firebrand'. Moreover, as Schindler saw (loc. cit.), Latin also shows isolated *i*-stem abstracts that are the pendants to corresponding thematic adjectives elsewhere. Thus *febris* 'fever' reflects an abstract *d^heg^{wh}-*ri-* 'heat', beside the *o*-stem adjective *d^heg^{wh}-*ro-* 'hot, burning' that is attested only elsewhere, in this case indirectly (though clearly) by Gk. τέφρα 'ashes', originally a collective based on such an adjective.[2] Likewise, Latin *imber* 'rain'[3] is readily interpretable as a concretized abstract 'moisture, wetness' (*nb^h-*ri-*), beside the substantivized *ro*-adjective (originally meaning 'wet') that appears in Sanskrit *abhrá-* 'rain-cloud' (*nb^h-*ró-*) and perhaps also in Gk. ἀφρός 'foam'.

3. 1. 3.

Indeed, for Latin, such *i*-stem abstracts must have played a significant rôle in an early phase of the system of nominal derivation, as recently shown by

[1] According to Rasmussen (2002: 337f.), the alternation just described fundamentally correlates with accent: *-*i-* is in origin merely the regular unaccented variant of the thematic vowel, and has no special association with substantival function. But even if this was true for some remote stage of the protolanguage, the semantic correlation of these *i*-forms with substantives is a descriptive fact of reconstructable Proto-Indo-European that Rasmussen's counterarguments (also invoking Meissner 1998: 240ff. on "Caland -*i*-") are in my view incapable of undermining.

[2] I discuss elsewhere (2002) such full-grade collective formations in *-*ro-* and their relationship to adjectival *ro*-stems.

[3] Probably together with the Oscan theonym **anafríss** (twice in the Agnone tablet, [ST] Sa 1 A.9, B.12).

A. Nussbaum. In his contribution to the Gedenkschrift for Jochem Schindler (1999), Nussbaum demonstrated that the Latin adjectives in *-idus* can be understood as "adjectivizations" of precisely such *i*-abstracts; thus, for example, Lat. *rūbidus* 'red' is built, by means of the adjectival formant *-d^ho-,[1] on an *i*-abstract h_1roud^h-*i-* or h_1reud^h-*i-* (in Indo-European terms) with a meaning like 'redness', beside the more original thematic adjective h_1roud^h-*o-* or h_1reud^h-*o-* 'red' (as in dialectal Lat. *rūfus* 'red-headed', and in the Germanic adjective for 'red': Goth. *raups*, OE *rēod*, etc.). Moreover, the fact that there is a considerable number of relatively isolated *-idus* forms in Latin makes clear that such *i*-abstracts must have been fairly well-attested in the prehistory of Latin.

3. 2.

Before considering how this theory could be applied to the morphological problem surrounding Lat. *cīuis*, we must first review the form's semantic background.

3. 2. 1.

As seen at the outset (**1.1.3.**), forms like Skt. *śéva-* and comparable material in Germanic and Baltic point to an original *o*-stem adjective *$k̂e\underline{i}$-*u̯o-* (or, much less likely, *$k̂e\underline{i}H$-*u̯o-*, with root-final laryngeal [cf. p. 139 n. 1 above, and further discussion directly below, **3.2.1.1.**]). The semantic reconstruction of this form is less clear; but as often, a lucid and (in my view) largely persuasive account of the problem can be found in Benveniste's still indispensable study of the vocabulary of Indo-European institutions (1969: 335-7). Without recapitulating Benveniste's argumentation in full, we may recall some of the salient points of his analysis. The Germanic and Baltic material, with meanings like 'husband', 'wife', and 'household', shows a specialization toward relationships that bind the members of a familial group ("la situation des conjoints unis par le lien du mariage et au cercle de la famille", 336).[2] In a similar way, Skt. *śéva-* and *śivá-*, in the hymns of the Rig-Veda, take part in traditional phraseology in collocation with *sákhā-* 'friend, companion', therefore implying a relationship among members of a close group of individuals ("un certain type de comportement amical à l'égard des partenaires d'alliance", 337). Finally, Lat. *cīuis* more properly means (according to Benveniste) not 'citizen' ('citoyen'), but rather 'fellow-citizen' ('concitoyen');

[1] Or possibly the second compound member *-d^hh_1o-, according to Hackstein (2002: 17) and Balles (2003).

[2] Ignasi-Xavier Adiego Lajara points out to me that it may be possible to add to this group CLuv. *zīti-* 'man', Lyc. *sidei-* 'son-in-law (?)' (on the meaning, see Melchert 2004: 57f., s.v.) < *$k̂e\underline{i}$-*ti-* or *$k̂o\underline{i}$-*ti-* (Melchert 1994: 241); for the etymology (based on *$ke\underline{i}$- 'lie', on which see further **3.2.1.1.**, directly below), see Gusmani 1987/8: 109.

it is thus a designation referring, at least originally, to one of the members of a group that holds, so to speak, "native rights" in a particular place ("les membres d'un groupe détenteur des droits d'indigénat", 337). It is only in Latin, then, that we see a development towards what Benveniste describes as an "institutional" or "political" sense ("copartageant des droits politiques", 337); by contrast, the original sense of the reconstructed adjective in question would have been, more broadly, something like 'socially close', with reference to an "in-group".

3. 2. 1. 1.

We can conveniently return, at this point, to the question of which Indo-European root underlies Skt. *śéva-*, Lat. *cīuis*, and related forms (cf. **1.1.2.**). The traditional assumption (see recently e.g. Wachter 1998: 204 with n. 26) is that we have to do with a specialized meaning of the familiar verbal root **ĥeį-* 'lie' (LIV 320); but the semantic details of this account have never been satisfactorily explained,[1] and this theory can be abandoned with little hesitation. A far more attractive analysis has been proposed in a laconic footnote from the prolific pen of F. Bader (1982: 133n240), and indeed her proposal (thus far largely unknown) can be further specified and even strengthened. Bader characterizes Lat. *cīuis* as an "expression pronominale d'une relation sociale à caractère politique comportant réciprocité", and beneath her reconstructed form **kei-wi-* she identifies the **ke-* [sic] that underlies the diphthongal **ke-i* of Gk. ἐ-κεῖ 'there, in that place'. For **kei-* + **-wo-* she compares Skt. *śéva-*, noting as a parallel formation **koi-* + **-mo-* in OIr. *cóim* 'dear'.[2] Although this is far from transparent in Bader's treatment, her "expression pronominale" **kei-/*koi-* [sic] (or Lamberterie's "base pronominale **kei-*, **kai-*, **koi-*" [sic]) evidently refers to the PIE near-deictic ("ich-deiktische") particle that is reconstructable in the forms **ĥke* and **ĥi* (IEW 609f., Dunkel 1990, 2004a: 45ff., LIPP s.v. **ĥke*), as in Hitt. *ĥā* 'here', Lat. *cedo/cette* 'give (it) here!', Lith. *še* 'here', Hitt. *kinun* 'now', Lat. *cis* 'on this side', and many other forms, including the derived pronominal stem **ĥko-/*ĥki-* 'this (here)', as in Hitt. *ĥaš*, CLuv. *zās*, HLuv. *zas* 'this', Lith. *šis* 'this', etc. The semantic value, then, is not, strictly speaking, "reciprocal" (as in Bader's formulation), but rather *near-deictic* (i.e., that which is close, in some sense, to the speaker).[3] From a formal point of view, suffixation via **-ų̯o-* is best understood as involving the "oppositional" affix elucidated by Cardona (1987: 4, cf. Katz 1998a: 98f.) and well-attested in a number of forms based on pronominal oppositions (see

[1] See the eloquent comments by Leumann on this point (1933: 238).

[2] Along similar lines: C. de Lamberterie (1990: 845, with reference to Bader). W. P. Lehmann (1985) seems to have arrived, independently, at a similar conception.

[3] Bader's interpretation was based in part on her assumption (loc. cit.) that Lat. *reciprocus* also belongs with this material; but that form is better understood differently, see Dunkel 1979 and LIPP s.v. **pro.*

e.g. Dunkel 2003: 13, 2004b: 23 on Ved. *sárva-*, *víśva-*, *pū́rva-*, etc., including *śéva-*).[1] Indeed, a pronominal **ĥki-ų̯o-* is directly attested in Umbrian (adv. **çive** 'davor, diesseits, außerhalb', IIb 11; Untermann 2000: 399), and immediately explains the laryngeal-less zero grade of Skt. *śivá-* 'dear' (< *'close [from the speaker's point of view]'). Thus, the etymology and suffixal composition of Skt. *śivá/śéva-*, Lat. *cīuis*, and related forms provide important support for Benveniste's semantic analysis in terms of "social closeness".[2]

3. 2. 2.

Now according to the derivational theory developed by Schindler (**3.1.1.**), a thematic adjective **ĥkéių̯o-* 'socially close' could have had, as a regular pendant, an *i*-stem abstract **ĥkéiu̯i-*. This abstract would in the first instance have meant something like 'societal closeness', that is, 'society' in the abstract sense. An utterly natural (and perhaps almost automatic) development would have led to a concretized sense of 'society' – that is, 'society' in the sense of "a socially close group of people; the members of a society".

3. 3.

As a further preliminary, before we pursue the implications of this morphological framework, let us consider yet another semantic fact that in my view has been insufficiently appreciated. The Latin adjectives *cīuicus* and *cīuīlis* are universally assumed to be based on the stem of *cīuis* itself. But if one thinks especially of the traditional phraseology associated with these forms, some of which – like *iūs cīuīle* – appears to be very old (note also *rēs cīuīlēs*, *ratiō cīuīlis*, and even, with singular, *uir cīuīlis*), it becomes clear that *cīuicus* and *cīuīlis* do not actually mean 'pertaining to a (fellow) citizen' so much as 'pertaining to society/the civic order'. Or at best, these forms naturally incorporate both types of meaning, without any evidence that the reference to a single 'citizen' is more basic than the reference to 'society' in a broader sense. As further support for this position: note that

[1] Further e.g. **oĺ-ų̯o-* (Gk. οἶ(ϝ)ος), **pri-ų̯o-* (OCS *prĭvŭ*) and **preĺ-ų̯o-* (Lat. *prīuus*), **proti-ų̯o-* (OCS *protivŭ* < remodeled **protei-ų̯o-*), **deks(i)-ų̯o-* (Goth. *taihswa*, Gk. δεξι(ϝ)ός), among other such forms.

[2] It is an open question (to which I hope to return elsewhere) how best to interpret, in morphological terms, the full grade **keį-ų̯o-* of Skt. *śéva-* etc. (as opposed to the zero grade of *śivá-*), e.g. as a vṛddhi formation of some kind (under the assumption that a form like *śivá-* is more original), or as based on the locative sg. **kéį* (Gk. κεῖ, ἐκεῖ etc.), or in connection with the accentual properties of **-ų̯o-* itself. In any event, the analysis just outlined is easily extendable to various **ĥ(e/o)j-mo-* forms in the same semantic sphere (in addition to OIr. *cóim* [cf. above], note also words for 'home' and 'family' in Germanic and Balto-Slavic: Goth. *haims*, Latv. *sáime*, etc.), as Bader correctly saw: cf., again, deictic **ĥi-mo-* in Umbrian (adv. **çimu**/*śimo* 'back [here]'; Untermann 2000: 397f.), as well as the frequent use of **-mo-* in derived adjectives based on particles, e.g. **pro-mo-* (Gk. πρόμος, U. *promom*, Goth. *fram*), etc. The derivational pattern, in fact, extends still further: cf. **keį -ro-* in Arm. *sēr* 'love', and the *l*-formation in OIr. *céile* 'companion'.

in the language of the law, Cicero makes use of *cīuīlis* to translate Greek πολιτικός, again in a meaning referring to society at large (in other words, referring to the πόλις, and not the πολίτης).[1]

3. 4.

With these considerations as background, we can now turn to the specific proposal for Lat. *cīuis*.

3. 4. 1.

I suggest, first of all, that what Latin actually inherited was not an original thematic adjective **ḱéi̯u̯o-* (as in Skt. *śéva-* and comparable forms elsewhere), but rather its pendant i-stem abstract **ḱéi̯u̯i-* 'society'. Under this assumption, then, the i-stem inflection of Lat. *cīuis*, far from being unexpected, is simply normal and predicted. (Nor is it problematic that the thematic adjective **ḱéi̯u̯o-* itself is nowhere attested in Latin, given comparable situations as in *febris* and *imber* [3.1.2.].) The problem, then, is shifted: there is no longer a difficult morphological anomaly requiring an instance of "inflectional contamination"; instead, the question becomes essentially a semantic one – we must understand how the abstract (or rather concretized abstract) **ḱéi̯u̯i-* 'society' developed, so to speak, a corresponding "singulative" meaning, namely 'a member of society'.

3. 4. 2.

Indeed, this development can be viewed as almost automatic, in at least two ways. As soon as an abstract term like 'society' becomes concretized, so as to mean 'the members of (a) society', its "collective" (or "distributive") sense might well come to be expressed with a plural, ending up as Lat. *cīuēs* – which, in fact, can have precisely this meaning, that is, "the citizenry". Lat. *cīuis*, then, would simply be the automatically back-formed singular, needed to express 'a (single) member of society'. But an even simpler conception begins from the typologically common linguistic development (a subtype of the concretization of abstracts, itself a banal pattern) according to which abstract notions referring to human activity become concretized with reference to the group – and eventually a single human – engaged in the activity: hence, for example, English *guard* (abstract, as in "be on one's guard") vs. *guard* (concrete/collective, e.g. in the military sense) vs. *a guard* (a single member of a [collective] *guard*), and similar terms in other languages, including Latin.[2]

[1] Fin. 4.5: *eum locum in philosophia, quem ciuilem recte appellaturi uidemur, Graeci* πολιτικόν; cf. also Fin. 5.66 (Ernout-Meillet s.v. *cīuis*; TLL s.v. *cīuīlis*, III.1213.66).

[2] Like Eng. *guard*: also *watch* and German *Wache*, Spanish/Italian *guardia*, Lat. *vigilia*, etc. In a semantic area closer to the social sphere of Lat. *cīuis*, note colloquial Eng. *family* in the sense '(equivalent to) a member of the family' ("Don't be so formal – you're family!", or "It's OK, I'm family", etc.).

As for the common (m./f.) gender of *cīuis*, as against the unmarked expectation that i-stem abstracts of the type under discussion might surface as feminines (cf. the type *rauis* [f.] 'hoarseness', **3.1.2.**): there are two ways in which this could be accounted for.[1] The simplest assumption is that this feature was concomitant with (and a consequence of) the "personalization" of the abstract, at which point the word came to refer to animates, and for the most part males, given Roman societal norms. Comparable developments are well-attested, as in Lat. *optiō* (m.) 'adjutant, assistant' (*the one chosen* [e.g.] by the centurion; hence a concretization and "personalization" of abstract *optiō* [f.] 'choice'). A different type of account emerges from a recent study by Nussbaum (2004), who identifies two subtypes among the forms that we have been referring to more generally as "i-abstracts", namely (a) the familiar feminine abstracts themselves, and (b) masculine endocentric substantivizations (i.e. 'that which is X'). Hence, for example, the i-forms based on PIE **h₂ák̂ro-* 'high, pointed' (**3.1.1.**): the original feminine abstract (itself later substantivized) is continued in Gk. ἄκρις (f.), while the masculine endocentric substantivization ('that which is high, pointed') appears in Lat. *ocris* (m.). These considerations, in turn, allow for an alternative account of the semantics of *cīuis*, which can instead be viewed simply as the masculine endocentric i-stem ('that one who is socially close') derived from the thematic adjective **ḱéi̯u̯o-* 'socially close'.

3. 4. 3.

Under the general scenario described above, there would be no need to resort to influence from *hostis*. But this possibility may still be relevant, though in a different way from what has been supposed. Although this point has rarely been noted, we should rather consider the combined effect of the precursors of the adjectives *hosticus* and *hostīlis*, beside *cīuicus* and *cīuīlis*, rather than placing the full burden on *hostis* (beside *cīuis*) itself. Given these adjectival pairings, which oppose, in a general way, the "out-group" (*hosticus, hostīlis*) to the "in-group" (*cīuicus, cīuīlis*), it seems natural to assume that influence from *hostis*, which refers to a single member of the out-group, may have helped establish the new "singulative" sense ('member of the in-group society, (fellow) citizen') of the concretized abstract **ḱéi̯u̯i-* 'society' – a development that could be viewed as a kind of semantic "Konträrbildung" to *hostis*. At the same time, the familiar "collective" behavior of the plural *hostēs* 'the enemy' (for example, in a military sense) would have paralleled the original "collective" sense of *cīuēs*, noted above (**3.4.2.**) as a possibility

[1] I am grateful to Andreas Willi for pointing out to me the interest of this issue.

(that is, 'society' in the sense of 'the citizenry'), which would again lead naturally to a contrast between singular *hostis* 'an enemy' and a back-formed singular *cīuis* 'a citizen'. Thus *hostis* could well have played a rôle in helping to establish or reinforce the *semantics* of *cīuis*, although not necessarily its inflectional form.

3. 4. 4.

The version of the above proposal that operates with an original *i*-abstract invites the question of what happened to the old abstract meaning of **k̂éi̯u̯i-*, once it had developed its new singulative (or personalized) meaning. Part of the answer has already been given (3.3.): the older meaning may survive in the derived adjectives *cīuicus* and *cīuīlis*, where, as I have suggested, the general "societal" sense need not have arisen secondarily from *cīuis* 'citizen', but may instead be older. (It does need to be stressed that in derivational morphology, derived material often preserves more archaic semantics than the base form itself.) The rest of the answer, however, is that this older sense of **k̂éi̯u̯i-* might naturally have been renewed by forming a new abstract, using a productive Italic formant like **-tāt-*: the result of such a renewal would be pre-Lat. **k̂ei̯u̯i-tāt-* > Lat. *cīuitās*.[1] Once again, this word combines both a general "societal" meaning (that is, as a collective term referring to a grouping of citizens, essentially in a political sense), as well as a meaning that refers to the individual citizen (namely, the 'condition of citizenship', or, so to speak, 'citizen-hood'). It is usually assumed – on no particular basis, apart from the supposition that the word is derived from *cīuis* 'citizen' – that the "individualized" meaning of *cīuitās* is primary. But this need not be the case: it is easy to imagine, in fact, that a meaning of this sort could have developed secondarily by association not only with the new "singulative" meaning of *cīuis* itself, but also by association with forms like *lībertās* 'the condition of being *līber* (as an individual)' or *societās* 'the condition of being a *sociu̇s*' (cf. Ernout-Meillet s.v. *cīuis*).

3. 5.

It would be impossible, in the limited scope of such an article, to exhaust various other theoretical possibilities for explaining Lat. *cīuis*. But it may be worthwhile to sketch two other conceivable approaches, both of which seem to me problematic.

3. 5. 1.

There is a class of Latin adjectives in *-īlis* correlated with thematic nouns,

[1] Strictly speaking, Latin vowel-weakening makes it impossible to determine whether the form contains a thematic first member (**k̂ei̯u̯o-teh₂t-*) or an *i*-stem first member (**k̂ei̯u̯i-teh₂t-*). Since, however, the framework under discussion dispenses entirely with a pre-Latin thematic **k̂ei̯u̯o-*, it is most straightforward to assume that the original *i*-stem **k̂ei̯u̯i-* (or pre-Lat. **k̂ei̯u̯i-*) was simply renewed as **k̂ei̯u̯i-teh₂t-* (pre-Lat. **k̂ei̯u̯i-tāt-*).

of the type *seruos : seruīlis, erus : erīlis, uir : uirīlis*, etc.[1] In theory, one could claim that *cīuīlis* is an adjective of this type, originally correlated with an *o*-stem **cīuos*; then, under influence from *hostis : hostīlis*, the *o*-stem **cīuos* was remodeled to an *i*-stem *cīuis*. It is, nevertheless, problematic once again that no trace whatsoever of the original *o*-stem survives, especially since the pattern *seruos: seruīlis* etc. (which is typical precisely for designations of persons: *seruos, erus, uir, puer*, etc.; Leumann 1977: 350) is very well-established in Latin.

3. 5. 2.

One could also begin from an *i*-adjective **k̂éi̯u̯i-*, which, like the theory I have proposed, would involve direct descent from an *i*-stem. But these *i*-adjectives belong to a relatively rare and poorly-understood type, reflected in Latin, for example, by *lēuis* 'smooth' (beside Gk. λεῖος 'id.').[2] Under this sort of approach, however, it would be difficult to understand how such an *i*-adjective **k̂éi̯u̯i-* (presumably with the same meaning as that of the thematic adjective **k̂éi̯u̯o-*) would have been substantivized with a meaning like that of Latin *cīuis*; nor is it immediately clear why adjectives **kei̯u̯i-ko-* and **kei̯u̯ī-li-* would have been needed as replacements for the original adjective **k̂éi̯u̯i-*, unless the substantivization took place at an exceedingly early date. Still, this direct *i*-stem source may be worth considering further. (Cf. also 4.3.3. below, with a somewhat different conception of adjectival *i*-formations.)

4. Conclusions

4. 1.

The most general conclusion, as a point of linguistic theory and methodology, is that the above analysis points to the advisability of constraining traditional accounts that appeal to "inflectional contamination", a suspect and (at least in some cases) overly-simplistic concept that may be of limited usefulness in historical morphology.

4. 2.

More specifically, for Latin: it would be worthwhile to reexamine similar cases of unexpected or historically anomalous *i*-stem inflection. By way of example, let us first briefly consider Lat. *collis* (m.) 'hill'. It is fairly clear that

[1] Ultimately, such alternations display, in a somewhat different guise, the archaic derivational relationship **-o-* ~ **-i-* discussed above; cf. already Nussbaum 1975: 132ff.

[2] These and other such forms have recently been discussed by Nussbaum (1999: 401f.).

Lat. *collis* must reflect a form **kol[H]-ni-*, in Proto-Indo-European terms, and is therefore almost identical, except for its *i*-stem inflection, to Lith. *kálnas* 'mountain'. Both forms are based on the root **kelH-* 'to raise' (LIV 349) otherwise seen in Lat. *-cellere* (as in *excellere* 'to be eminent'), *celsus* 'lofty', *columen/culmen* 'top, summit', and various other forms elsewhere (Li. *kélti* 'to raise', OE *hyll* [Eng. *hill*], etc.; IEW 1. *kel-*, 544), including, in my opinion, South Picene **qolofítúr**, as I discuss elsewhere (Vine, forthcoming).[1] Now the *i*-inflection of Lat. *collis* has never been explained; it is simply listed, for example, by Leumann (1977: 343), among other such forms (including *cīuis*), without comment. It would not be difficult, however, to imagine a solution that is almost exactly parallel to the one I have proposed for *cīuis*. Lith. *kálnas* is almost certain to derive from a substantivized adjective **kolH-no-* 'high'.[2] All that is needed to explain Lat. *collis* is the corresponding *i*-abstract **kolH-ni-* 'elevation, height' (or endocentric substantivization 'that which is high' [**3.4.2.**]; cf. the masc. gender), which was either concretized directly to mean 'a height, a high place', or which, in its collective sense as '(the) heights, high places', developed as a plural, from which a back-formed "singulative" *collis* 'an elevation, hill' was easily formed.[3]

4. 3.

Similar explanations likewise present themselves for other hitherto mysterious cases of *i*-inflection in Latin, such as the following.

4. 3. 1.

Semantically similar to *collis* is Lat. *mōns* (m.) 'mountain', perhaps best considered to continue a *ti*-stem **món-ti-*, despite the fact that (as I discuss elsewhere, 2004: 374ff.) *o*-grade in canonic *ti*-stems (i.e., *nomina actionis* built to verbal roots, in this case the **men-* 'project, tower over' [LIV 3. ****men-***] of Lat. *ē-/prō-minēre*) is highly anomalous.[4] Here we find evidence for thematic *t*-formations, as in Lat. *mentum* 'chin' (< *'prominence'), cf. OHG *mund* 'mouth' and other such forms (IEW 726). These forms could easily be substantivizations of an adjectival **mn̥-to-* 'projecting, jutting out', whence a parallel *i*-abstract **mónti-* 'prominence, projection',[1] concretized in Latin ('a projection, height' > 'mountain') in essentially the same fashion as with *collis* above.[2] In this way, incidentally, the otherwise obscure *o*-grade of *mōns* would receive a natural explanation.

[1] I leave aside here the question of the non-vocalization of the laryngeal in Lat. *collis*, which is a regular phonological development due to so-called "Saussure's Law" or "the Saussure effect" (Schrijver 1991: 326f., Nussbaum 1997: 195).

[2] The apparent Saussure's Law violation (cf. preceding footnote) can have an analogical explanation; cf., further, the parallel *yā*-formation Lith. *kalvà* (acc. *kalvą* etc.) 'high ground, hill', with expected circumflex intonation (Nussbaum loc. cit.).

[3] The derivational patterns just outlined may belong to a relatively late stage of the protolanguage (and even, for some forms, the individual languages/branches in question), if the ultimate source of such *n*-formations to this root is actually an athematic *n*-stem, as has often been suggested. See most recently Neri 2003: 273n904 (proposing a basis in an amphikinetic *n*-stem **kélH-ōn*).

[4] According to Untermann (1992: 144 with n. 23), *mōns* is better viewed as an old *t*-stem, in view of its derivative *montānus*; but such forms can be entirely secondary, cf. *urbānus* based on the *i*-stem *urbi-* of *urbs* (Leumann 1977: 324).

4. 3. 2.

Lat. *testis* (m./f.) 'witness' beside the thematic **testo-* implied by the denominative *testārī* (cf. Oscan **trstus, trístaamentud**) has been tentatively explained as another contamination result.[3] To consider the vocable (anachronistically, but conveniently) in its Latin form,[4] we could instead imagine an original thematic adjectival **testo-* 'standing as witness' beside a derived *i*-abstract **testi-* 'witnessing' (or an endocentric substantivization 'the witnessing one'), whence a meaning '(individual) witness', much as in Eng. abstract *witness* (cf. usages like "bear witness") vs. concrete *witness* ("a person who bears witness"), similarly French *témoin*, etc.

4. 3. 3.

According to the theory developed by Nussbaum (2004), masculine endocentric derivatives of the type Lat. *ocris* (**3.4.2.**), **monti-* (**4.3.1.** with n. 2), etc. show a tendency to develop further as "weak adjectives" – and indeed, this provides, for Nussbaum, the explanation for the *i*-stem inflection of Lat. *tenuis* 'thin' (originally 'that which is thin, thin (one)' < **tenh₂u̯i-*) beside Gk. ταναός 'id.' (< **tn̥h₂u̯o-*). If this is so, a comparable account could perhaps be provided for Lat. *fortis* (**1.2.2.**), which could in principle reflect an original endocentric derivative ('that which is strong, strong (one)' > adj. 'strong') beside an original thematic formation, perhaps continued in Lat. *horctum/forctum*.

4. 4.

Naturally, there is a residue of difficult forms, for which analyses of this sort do not appear to be demonstrable, and for which further research is needed. Thus, for example, Lat. *caulis* (m.) 'stem (of a plant)', which belongs with

[1] Perhaps indirectly attested in YAv. *maiti-* (f.) 'id.' < **mn̥t-i-* (cf. Tremblay 1999: 132), with the zero grade suggesting that the form is decompositional (cf. the type Skt. *rudhi*^{o} beside the **h₁reu̯dʰ-i-*/**h₁rēu̯dʰ-i-* 'redness' discussed above, **3.1.3.**).

[2] Much the same account of this material has recently been presented by Nussbaum (2004), who treats Lat. **monti-* (m.) as an endocentric substantivization ('that which projects'), of the type discussed above (**3.4.2.**).

[3] I.e., *testis* after *orbis* (Katz 1998b: 209n69, citing *cīuis* after *hostis*).

[4] On the etymological details, see Katz 1998b: 184ff.

Gk. καυλός 'stem, pole', Lith. *káulas* 'bone', and OIr. (*ā*-stem) *cúal* 'faggot, bundle of sticks' (IEW 537), is ordinarily assumed to have developed its *i*-stem inflection secondarily (thus e.g. Frisk GEW I: 803, Leumann 1977: 343, both without explanation). It is in principle possible that the Greek, Lithuanian, and Celtic forms reflect independent substantivizations of an original adjectival **keh₂u-ló-*, beside a corresponding concretized *i*-abstract (or rather, given its gender, an endocentric substantivization) in Latin.[1] But further analysis is hampered by our ignorance about the root and its meaning.

Still more difficult is Lat. *piscis* (m.) 'fish' (not easy to derive from an abstract notion), beside not only the thematic formation implied by the Latin denominative verb *piscārī* (cf. also *piscātor* 'fisherman', and in general the pattern seen in *testis* : *testārī*, **4.3.2.**), but also *o*-stem forms elsewhere, in the same meaning, as in Germanic (Goth. *fisks*, OIc. *fiskr* etc., from a proximate **pisko-*) and Celtic (OIr. *íasc*, as if from a full grade **peisko-*).[2]

4. 5.

Finally, with regard to Lat. *cīuis* and *cīuitās* themselves: if, as I have argued, Lat. *cīuis* actually continues not a "contaminated" version of the *o*-stem adjective **ĥéi̯uo-* seen in Skt. *śéva-* and similar forms, but rather a regularly-derived abstract (of the type *febris, imber*, cf. also *collis, mōns, testis*), then we have uncovered, in pre-Lat. **kei̯ui̯-*, a small but potentially interesting new fragment of the social and political lexicon inherited by Latin. Moreover, the existence at some early stage of Latin of such an abstract or substantivization with a meaning like 'society' (later replaced with **kei̯ui̯-tāt-*, **3.4.4.**) might help explain the otherwise puzzling absence, in Latin as compared with Sabellian, of the old term **teu̯tā-*. But any such further ramifications of this material must be left to specialists in Italic culture.[3]

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[1] For the reconstruction **keh₂u-ló-*, see Schrijver 1991: 268f.

[2] Perhaps the *i*-stem *piscis* was back-formed on the basis of *piscīna* 'fish pond' [Plt.+], which is synchronically ambiguous between *o*-stem (*equo-/equīno-*) and *i*-stem (*colli-/collīno-*) sources.

[3] Many participants of the *Secondo Convegno Internazionale sugli Antichi Umbri* ("La città italica"; Gubbio, September 2003), where this paper was first presented, kindly offered constructive criticism and helpful comments, and similar benefits were conferred by audience members of the panel on Greek and Latin linguistics at the *135th Annual Meeting of the American Philological Association* (January, 2004), who heard another version of this material; to all I extend my warmest thanks.

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